1. PHASE ALPHA: CRISIS HYDRATION ("OMG I NEED WATER NOW!!!")

■ **THE CONTAMINATION ASSUMPTION:** Treat all open surface water (creeks, rivers, drainage ditches, standing puddles, roof runoff) as biologically lethal. Clear water does NOT mean safe water. Never drink a single drop of unverified water under any circumstance.

■ **THE RAPID SILT SEPARATION:** Pour raw water through a clean cotton shirt, bandana, or paper coffee filter stretched over a container mouth. This stops heavy mud, sand, and leaves from fouling your secondary treatment layers.

■ **IMMEDIATE CRITICAL DROPS:** If you have standard liquid chlorine bleach but no power to boil, immediately locate a clean container. Add exactly 8 drops of regular, unscented 5.25% to 8.25% liquid bleach per gallon of clear pre-filtered water. Shake for 10 seconds. Loose-thread the cap to let the liquid rinse the neck threads, tighten, and wait a mandatory 30 minutes before drinking.

■ **CRITICAL ROLLING BOIL:** If a heat source is active, transfer pre-filtered water to a metal cup or pot. Bring the liquid to a violent, active rolling boil (large breaking bubbles continuously bursting across the surface) for 60 seconds. Let the vessel cool completely before opening.

CHEMICAL DISINFECTION REGISTRY

WATER CONDITION	BLEACH INJECTION VOLUME	MANDATORY CONTACT TIME
Clear / Pre-filtered	8 Drops Per Gallon (0.5 mL)	30 Minutes (Water temp > 50°F)
Cloudy / Turbid / Cold	16 Drops Per Gallon (1.0 mL)	60 Minutes (Water temp < 50°F)
Bulk Storage (55 gal drum)	2 Tablespoons (1 Fluid Ounce)	24 Hours prior to dynamic entry

2. PHASE BRAVO: LONG-TERM WATER LOGISTICS & MECHANICAL PURIFICATION

■ **THE MULTI-TIER GRAVITY FILTER ASSEMBLY:** When chemical stocks deplete, construct a survival filter column to process turbid surface water. Pierce five 1/16-inch holes into the bottom cap of a clean 5-gallon plastic drum or inverted soda bottle. Layer materials from the bottom upward in this exact sequence:

1. Bottom Layer: 2 inches of tightly compressed clean cloth or medical gauze pads.
2. Second Layer: 4 inches of pulverized hardwood charcoal crushed into a fine sand consistency (traps chemical toxins).
3. Third Layer: 6 inches of clean, washed fine river sand.
4. Top Layer: 3 inches of clean pea gravel (strips out heavy twigs and organic debris).

■ **OPERATIONAL LOOP:** Pour raw water into the top gravel layer. Collect the clear effluent from the bottom holes. CRITICAL ENFORCEMENT: This process only removes sediment and odors. You MUST still execute a thermal boil or chemical drops cycle on the collected clear water to kill microscopic pathogens.

■ **FLOCCULATION METHOD:** If surface water is thick with mud or suspended clay particles, dissolve 1/4 teaspoon of alum (potassium aluminum sulfate) per gallon of water. Stir rapidly for 1 minute, then slowly for 5 minutes. Let it stand undisturbed for 60 minutes. The mud will lock together and sink to the absolute bottom of the vessel as heavy sediment flakes, allowing you to siphon clean water from the top.

3. POTABLE WATER CUSHION AUDIT LOG

CACHE LOCATION INDEX	TOTAL GALLONS ON HAND	CHEMICAL ROTATION DATE	VERIFIED POTABLE (■)
Primary Storage Rack	_____	_____	[ ]
Vehicle 1 Trunk Cache	_____	_____	[ ]
Secondary Safe Vault	_____	_____	[ ]

CRITICAL WATER SECURITY & PUBLIC HEALTH PANEL
AUTHORITATIVE REFERENCES: U.S. Environmental Protection Agency Guidelines Manual (EPA-816-F-15-003), Centers for Disease Control and Prevention (CDC) Emergency Hydration Standards.

LEGAL DISCLAIMER: Private processing estimates do not supersede municipal boil-water notices or official contamination directives issued by county health officers. Always check container labels to confirm exact concentration strings.